SmartRoad BD

AI-Powered Road Infrastructure Intelligence System

Software Project Proposal

Department of Computer Science and Engineering

Problem Domain & Background

Bangladesh is one of the most densely populated countries in South Asia, with over 170 million citizens sharing an urban infrastructure that was never designed to handle current demands. In major cities like Dhaka, Chittagong, Sylhet, and Rajshahi, road infrastructure has become a critical failure point in the daily lives of millions of people. Every year, rapid urbanization, population growth, and unchecked vehicle density place enormous and unrelenting pressure on road surfaces that are already fragile, underrepresented in maintenance budgets, and largely invisible to any systematic oversight.

The structural problem is not simply one of physical decay. It is a failure of information, coordination, and accountability across the entire infrastructure management chain. Roads deteriorate silently. Potholes deepen unnoticed. Cracks widen into subsidence. Surface wear exposes underlying layers to monsoon flooding. And throughout all of this, citizens, city authorities, and field maintenance teams remain disconnected — each group lacking the information and tools they need to act effectively.

To address this multi-layered crisis, the proposed system SmartRoad BD is designed as a complete, AI-driven road infrastructure intelligence ecosystem. It connects citizens who observe damage, data systems that analyze it, government authorities who must prioritize and fund repairs, and field engineers who execute them — within a single, closed, accountable loop. SmartRoad BD is not a complaint box. It is a living infrastructure intelligence platform that transforms how Bangladesh monitors, maintains, and manages its road network.

Root Cause of the Problem

The deterioration of Bangladesh's urban road infrastructure is not a sudden occurrence. It is the cumulative result of years of neglect, poor institutional coordination, and a complete absence of digital management tools designed for the reality of road maintenance in a rapidly growing city. The root causes are systemic, interconnected, and mutually reinforcing.

The specific root causes of this problem are:

- No centralized road damage reporting system for citizens
- Absence of AI-driven damage detection and severity classification
- No data-driven mechanism for prioritizing road repairs across competing needs
- Zero accountability structure between reported damage and completed repair
- Manual and fragmented field team coordination with no digital task assignment
- Reactive-only maintenance culture — roads are fixed only after visible failure, not before
- No predictive degradation modeling to forecast which roads will fail next
- No seasonal vulnerability mapping despite predictable monsoon damage every year
- Low citizen engagement due to no feedback or acknowledgment of reports
- Duplicate and unverified complaint data flooding government offices without prioritization

- No budget optimization framework to direct limited public funds toward highest-impact repairs
- Lack of before-and-after repair verification allowing corruption and incomplete work

Importance of the Problem

The failure of road infrastructure management in Bangladesh is far more serious than a matter of everyday inconvenience. When examined at scale — across millions of daily commuters, thousands of road segments, and hundreds of maintenance decisions made without data — it becomes a crisis with measurable human, economic, and institutional consequences.

The specific reasons why this problem demands urgent, systematic attention include:

- **Public Safety Risk:** Every day, thousands of vehicles — from rickshaws to heavy trucks — strike hidden potholes and damaged road surfaces. These impacts cause direct vehicle damage, sudden accidents, and in severe cases, loss of human life. Pedestrians and cyclists are especially vulnerable in poorly lit conditions where road damage is invisible until it is too late.
- **Economic Productivity Loss:** Vehicle repairs from pothole damage, increased fuel consumption on rough roads, cargo losses from poorly maintained freight routes, and the time cost of traffic slowdowns caused by road damage collectively impose a massive hidden tax on Bangladesh's economy. Small businesses and daily wage earners bear a disproportionate share of this burden.
- **Emergency Response Delays:** Emergency vehicles — ambulances, fire trucks, and police response units — are slowed and sometimes unable to navigate critically damaged urban roads during emergencies. In a country where response time is already strained by traffic density, further delay caused by avoidable road damage directly costs lives.
- **Seasonal Damage Cycles:** Bangladesh's monsoon season arrives predictably every year, yet roads that flooded and broke apart last year are left equally vulnerable the next. Without a system that maps seasonal risk zones and triggers preemptive maintenance, the same damage recurs annually in the same locations, draining public budgets on avoidable repairs.
- **Erosion of Public Trust:** When citizens report potholes to Facebook pages, call local offices, or flag issues to ward commissioners — and receive no acknowledgment, no tracking, and no resolution — they lose trust in government institutions. This erosion of civic confidence undermines the social contract between the state and its citizens and reduces future cooperation on civic initiatives.
- **Misallocation of Public Resources:** Without a structured reporting system, maintenance decisions are made based on political visibility, personal connections, or random prioritization rather than objective evidence of severity or risk. This means the most dangerous roads are often not the ones repaired first, and limited government budgets are systematically misallocated.

Proposed Solution

SmartRoad BD is a complete, AI-powered road infrastructure intelligence system that connects citizens, data, government authorities, and field maintenance teams within a single, transparent, and accountable digital ecosystem. Unlike a conventional complaint portal, SmartRoad BD operates as a closed feedback loop — every report is analyzed, verified, prioritized, assigned, executed, and confirmed, with status updates returned to the original reporter at each stage.

The system operates across four integrated layers that work together to transform road management from a reactive, manual process into a proactive, data-driven operation.

Layer 1: Smart Citizen Reporting

Citizens report road damage through a mobile application with a single photograph. The app guides the camera angle for optimal capture quality, applies AI-based image enhancement for low-light conditions, and automatically captures the GPS coordinates, timestamp, and road segment identifier without any manual input from the user. This removes all barriers to reporting and ensures that every submission contains the spatial metadata required for downstream processing.

Layer 2: AI Analysis and Crowd Validation

Upon submission, a computer vision engine immediately classifies the damage type — pothole, surface crack, road subsidence, or waterlogging — and assigns a severity score from 0 to 100 using a weighted model that combines AI confidence, road traffic volume, historical accident data on that segment, and subsequent crowd validation votes. Nearby citizens are notified to confirm or dispute the report with additional photographs, building a community-verified trust score. Duplicate reports from the same 50-meter radius are automatically merged rather than listed separately, so volume becomes a meaningful signal of urgency rather than system noise.

Layer 3: Government Intelligence and Task Assignment

All verified reports feed into a real-time government authority dashboard that presents a severity-ranked list of active road issues across the city — not a pile of raw complaints. Each issue displays its composite severity score, traffic importance of the affected road, number of independent reports, estimated economic impact, and recommended repair materials. Authorities assign tasks to field engineer teams digitally, with a full log of assignment time, responsible officer, and team details. Field engineers receive a mobile notification with the exact GPS location, damage photographs, materials list, and an AI-optimized travel route to the repair site.

Layer 4: Verified Repair and Closed Feedback

After completing a repair, the field engineer must upload a geotagged post-repair photograph at the precise GPS coordinates of the original report. The system runs an AI-based before-and-after image comparison to verify that the damage has been genuinely addressed. Only upon passing verification is the repair marked complete. The original reporting citizen then receives a push notification confirming resolution, along with a prompt to rate the repair quality. This feedback is stored and contributes to contractor performance scores over time, creating an accountability system for repair quality that did not previously exist.

Predictive Intelligence Layer

Beyond reactive reporting and repair, SmartRoad BD incorporates a predictive degradation engine that forecasts infrastructure failure before it occurs. A machine learning model trained on road material age, historical repair frequency, traffic load, rainfall data, and soil composition scores every road segment monthly with a 90-day failure probability. Segments exceeding a risk threshold appear in the government dashboard's preventive maintenance queue even when no citizen report has been filed — shifting the system from reactive to proactive maintenance. A dedicated monsoon vulnerability map, generated each April before the rainy season, identifies the highest-risk segments to enable preemptive patching before annual flood damage begins.

This solution is particularly appropriate because it directly addresses every root cause simultaneously: it creates a reporting system, validates data quality, provides prioritization logic, enables digital field coordination, ensures repair accountability, incentivizes citizen engagement, and generates the historical data needed for long-term infrastructure planning. The system is fully feasible using established technologies and can be deployed incrementally beginning with a single city ward as a pilot before citywide rollout.

Target Group of Users

SmartRoad BD is designed to serve five distinct stakeholder groups, each with a dedicated interface tailored to their role within the road infrastructure management ecosystem.

- **General Citizens:** General citizens and daily commuters in Dhaka and other major cities who experience road damage firsthand and currently have no effective channel to report issues and receive follow-up.
- **City Authority Officials:** Municipal commissioners, city corporation departments, and ward-level officials who must make daily decisions about repair prioritization, budget allocation, and team deployment without access to structured data.
- **Field Maintenance Engineers:** Government-employed and contracted road maintenance teams who receive unclear instructions, lack GPS routing to damage sites, and currently have no digital system for logging or verifying completed work.
- **Platform Administrators:** System administrators responsible for managing user accounts, AI model performance, data integrity, and platform configuration across the full stakeholder ecosystem.
- **Researchers and Urban Planners:** Urban planning researchers, civil society organizations, NGOs, and national policy bodies that require structured, historical infrastructure data for evidence-based planning and advocacy.

Benefits of Target Users

For Citizens

- Report road damage in seconds with a single photo, without needing to know who to call or visit any office
- Receive real-time status updates at every stage — acknowledged, assigned, in repair, and verified complete
- Earn reward points for verified reports that can be redeemed via mobile banking or service discounts
- Access the live road heatmap to choose safer routes and receive push alerts about critical road damage nearby
- Voice-based and SMS reporting options serve low-literacy and low-connectivity users equally

For City Authorities

- Replace scattered, unverified complaint piles with a severity-ranked, data-backed priority list updated in real time
- Make budget decisions supported by cost-per-repair analytics, ROI projections, and resource allocation recommendations
- Access contractor performance scores and before-after repair verification to enforce accountability without field inspections
- Use the predictive maintenance queue to schedule preventive repairs before failures occur, reducing emergency spend
- Generate exportable reports for government audits, donor reporting, and public accountability disclosures

For Field Engineers

- Receive digital task assignments with complete site information: GPS location, damage photos, material requirements, and priority level
- Follow AI-optimized multi-site routing that minimizes total travel time across the day's repair assignments
- Submit geotagged before-and-after photos directly from the field app, eliminating paperwork and manual reporting
- Track personal task completion history and performance metrics for appraisal and professional development

Features of the Proposed Solution

Citizen Portal Features

1. **User Registration and Profile Management:** Citizens register with NID or phone number. Profile stores reporting history, earned points, saved locations, preferred language, and notification preferences across all sessions.

2. **AI-Guided Photo Reporting:** The app's camera interface provides real-time guidance on angle and distance for optimal AI analysis quality. Image enhancement algorithms process low-light and blurred inputs automatically before submission.

3. **Automatic GPS Tagging:** Every report is geo-stamped at the moment of submission. The system maps the coordinates to the nearest official road segment identifier from city GIS data, ensuring location accuracy without user input.

4. **Real-Time Road Heatmap:** A live, color-coded map displays all active road damage across the city. Red indicates critical danger zones, yellow indicates moderate damage, and green indicates recently verified safe roads. Citizens can filter by area or damage type.

5. **Smart Route Alerts:** Push notifications warn citizens when a high-severity road issue is detected on their frequently traveled routes. In-app route suggestions automatically route around dangerous segments during peak damage periods.

6. **Crowd Validation Participation:** Citizens within 500 meters of a new report receive validation requests. They can confirm, dispute, or add supplementary photos. Participation earns reward points and improves the quality of data in the system.

7. **Report Status Tracking:** Every submitted report has a live status tracker: Submitted, AI Analyzed, Community Verified, Assigned to Team, Repair In Progress, Repair Verified, and Closed. Citizens receive push notifications at each transition.

8. **Gamification and Rewards System:** Points are awarded for each verified report, crowd validation contribution, and quality rating submission. Monthly leaderboards rank top contributors by ward and city. Points are redeemable via bKash mobile banking integration.

9. **Multi-Language Support:** The full citizen interface is available in Bangla and English, with automatic language detection based on device settings. Voice-based reporting in Bangla is available for low-literacy users.

10. **SMS Fallback Reporting:** Citizens with limited smartphone capability or poor internet connectivity can submit reports via SMS with an attached MMS photo. The backend parses incoming messages, extracts location from the nearest cell tower data, and creates a report entry automatically.

Government Authority Dashboard Features

1. **Severity-Ranked Priority Dashboard:** The authority homepage displays all active road issues ranked by composite severity score, not chronological order. Each card shows the damage category, road name, severity score, number of reports, traffic volume tier, and estimated repair cost — enabling instant, evidence-based triage.

2. **Smart Task Assignment System:** Authorities assign repair tasks to field teams through the dashboard with a single click. The system logs the assigning officer, timestamp, target completion date, and links the task to the specific damage report with all associated evidence.

3. **Predictive Maintenance Queue:** A dedicated section of the dashboard lists road segments that have not yet been reported by citizens but are flagged by the AI degradation model as having above-threshold failure probability within 90 days, enabling preemptive action.

4. **Budget Optimization Module:** The analytics engine calculates cost-per-repair, repair recurrence rates, and estimated economic impact avoided per repair completed. Budget allocation recommendations are generated monthly by ward and road classification.

5. **Contractor Accountability Tracker:** Each contractor's performance is scored based on repair quality ratings, time-to-completion, and recurrence rate — defined as the rate at which roads they repaired are re-reported within 90 days. Low-scoring contractors are flagged for review.

6. **Historical Data Analytics:** Multi-year trend dashboards show damage frequency by location, seasonal patterns, repair completion rates, and response time averages. All data is exportable for government reporting and donor documentation.

7. **Monsoon Vulnerability Map:** Generated every April, this pre-season map overlays road drainage quality, elevation, soil permeability, and historical flood data to identify the segments most likely to sustain monsoon damage, with a recommended preemptive maintenance list.

8. **Before-After Verification Review:** For repairs that fail the AI auto-verification threshold, authorities can review side-by-side before-and-after photos and manually approve or reject the completion claim, providing a final human check in the accountability chain.

Field Engineer Application Features

1. **Digital Task List with Full Site Brief:** Engineers see their daily assigned tasks with complete information for each: GPS coordinates, damage category, severity score, pre-repair photos, recommended materials, and estimated repair time — all accessible offline after initial sync.

2. **AI-Optimized Multi-Site Routing:** When multiple repairs are assigned for a given day, the system calculates the optimal sequence and navigation route to minimize total travel time and fuel consumption across all sites.

3. **Geotagged Before-and-After Verification:** The field app enforces photo submission only at the GPS coordinates of the original report, preventing off-site photo uploads. Before-and-after pairs are automatically transmitted for AI verification upon job completion.

4. **Offline Functionality:** The full engineer workflow — viewing tasks, capturing photos, and updating status — functions without internet connectivity. Data syncs automatically when connection is restored, supporting engineers in areas with poor network coverage.

Core AI and System Features

1. **Computer Vision Damage Classification:** A convolutional neural network trained on a labeled dataset of Bangladeshi road damage images classifies each report into one of four primary categories — pothole, surface crack, road subsidence, and waterlogging — with confidence score and pixel-level damage extent estimation.

2. **Composite Severity Scoring Engine:** The final severity score (0–100) is calculated as a weighted combination: AI classification confidence (40%), crowd validation votes (20%), traffic volume of the road segment from open map data (25%), and historical accident density at that location (15%). This score is the sole basis for repair queue ranking.

3. **Duplicate Report Merging:** Reports submitted within 50 meters of an existing open issue are automatically grouped under the same incident record. Each additional report increases the issue's crowd weight score, so volume becomes a meaningful indicator of public concern rather than database noise.

4. **Predictive Degradation AI Model:** A gradient-boosted regression model trained on road material age, repair history, traffic load estimates, cumulative rainfall data, and soil type scores every mapped road segment monthly with a 0–100 failure probability for the next 90 days.

5. **Before-After Repair Verification:** A structural similarity model compares the pre-repair and post-repair images for each completed job at the confirmed GPS location. Jobs scoring above the repair confidence threshold are auto-approved. Those below threshold are escalated to authority review.

6. **Open Data API for Researchers:** Anonymized, aggregated infrastructure data is accessible via a documented REST API for registered academic researchers, NGOs, and urban planning organizations, supporting evidence-based advocacy and long-term policy development.

System Overview

The following table summarizes the end-to-end operational flow of SmartRoad BD across all stakeholder roles:

Stage	Actor	System Action	Output
1. Damage Report	Citizen	AI-guided photo capture + auto GPS tagging	Structured damage record
2. AI Analysis	AI Engine	Damage classification + severity scoring	Severity score 0–100
3. Crowd Validation	Nearby Citizens	Confirm / dispute / add photos	Trust-verified report
4. Map Update	System	Live heatmap updated + route alerts sent	Real-time infrastructure map
5. Priority Ranking	System	Composite score ranking across all issues	Ranked authority dashboard
6. Task Assignment	Authority	Digital task sent to field engineer	Assigned repair order
7. Repair Execution	Field Engineer	Navigate → repair → submit geotagged photos	Completion with evidence
8. AI Verification	AI Engine	Before-after image comparison	Verified repair record
9. Citizen Feedback	Citizen	Notification + quality rating submission	Closed feedback loop

Sustainability Model

SmartRoad BD is designed with long-term financial viability as a core requirement, not an afterthought. The platform is structured around four complementary revenue and funding streams that together ensure operational sustainability without dependence on any single source.

- **Government SaaS Subscription:** City corporations, municipalities, and district-level government bodies subscribe to the authority dashboard and field engineer application under an annual SaaS licensing model, scaled by city population and number of active road segments under management.
- **Smart City and Development Grants:** SmartRoad BD is eligible for smart city infrastructure funding from multilateral development organizations including the World Bank, Asian Development Bank, and UNDP, which actively fund digital governance and urban resilience projects in South Asia.
- **Data Analytics Services:** Urban planning institutes, infrastructure research centers, transport ministries, and international development organizations pay for structured access to anonymized, historical road degradation datasets through the open data API.
- **NGO and Donor Partnerships:** Road safety organizations, environmental NGOs, and international development partners co-fund the platform in exchange for priority access to policy-relevant insights and participation in platform governance.

Evaluation Questionnaire

Question 1: Does the team demonstrate a thorough understanding of the need, problem, or opportunity, including evidence of research into the need, problem, or opportunity?

Answer: Bangladesh has no centralized, digital road infrastructure monitoring system across its urban centers. In Dhaka alone, the Roads and Highways Department manages over 21,000 kilometers of classified roads, yet damage reporting relies entirely on informal channels — social media posts, verbal complaints to ward commissioners, and unstructured calls to government offices. There is no mechanism to capture, classify, or track these reports systematically. Citizens who report damage receive no acknowledgment. Authorities who receive complaints have no way to verify severity or prioritize response. Field engineers who carry out repairs have no digital workflow or verification system. This creates a complete absence of the intelligent feedback loop that modern infrastructure management requires. The consequences are measurable: road accidents in Bangladesh cause approximately 25,000 deaths and over 1 million injuries annually, with road surface conditions cited as a leading contributing factor. Seasonal monsoon flooding repeatedly destroys the same road segments year after year, demonstrating that the problem is predictable but unaddressed at the systemic level. SmartRoad BD is designed specifically around these documented realities, and its architecture reflects a deep understanding of the technical, behavioral, and institutional constraints that any solution in this context must navigate.

Question 2: Does the project have a clear objective including relevant benefits and target market or audience of the product?

Answer: SmartRoad BD has a precisely defined objective: to create a complete digital intelligence loop connecting citizen-reported road damage, AI-powered analysis, government priority management, and verified field repair — in real time, at scale, across Bangladesh's urban road network. The target users span five clearly defined stakeholder groups: general citizens and daily commuters who report and validate damage; city authority officials who prioritize and assign repair work; field maintenance engineers who execute and document repairs; platform administrators who manage system integrity; and researchers and urban planners who use historical data for policy development. Each group receives a dedicated, role-appropriate interface optimized for their specific workflow and context. Citizens benefit from a functioning voice in infrastructure management for the first time, a clear feedback loop on their contributions, and tangible safety improvements on their daily routes. Authorities benefit from structured, evidence-based decision support that replaces politically driven or arbitrary prioritization. Field engineers benefit from digital task management that reduces confusion, improves accountability, and simplifies their daily workflow. Researchers and planners benefit from access to high-quality, longitudinal infrastructure data that has never previously existed in Bangladesh at this granularity. The system directly improves public safety, reduces government waste, increases civic trust, and generates the data infrastructure Bangladesh needs for long-term smart city development.

Question 3: What are the solutions you are going to propose to deal with the problem? Why is this solution particularly appropriate to solve the problem? Is the solution feasible to meet the business objective?

Answer: SmartRoad BD proposes a multi-layered intelligent platform built on four integrated operational pillars. The first is a mobile-first citizen reporting system with AI-guided photo capture, automatic GPS tagging, and image enhancement for low-light conditions — removing every barrier between a citizen observing damage and that damage entering the system as structured data. The second is a dual-validation engine combining computer vision classification, composite severity scoring, and crowd-sourced community verification — ensuring that only trustworthy, categorized data reaches government decision-makers. The third is a government intelligence dashboard with severity-ranked repair queues, predictive maintenance alerts, budget optimization analytics, and contractor accountability tracking — replacing manual, politically influenced prioritization with data-driven governance. The fourth is a field engineer mobile application with GPS-guided task management, optimized multi-site routing, and mandatory geotagged before-and-after repair verification — creating a complete digital audit trail for every repair in the system. This solution is particularly appropriate because it addresses every identified root cause simultaneously and does not require behavioral change from citizens beyond taking a photograph. It is fully feasible using established and proven technologies: React Native and Flutter for mobile development, Python with TensorFlow or PyTorch for the AI components, Node.js and PostgreSQL with PostGIS for the backend and spatial database, and cloud infrastructure on AWS or Azure for scalability. All components are within the development capability of a skilled software engineering team and can be piloted at ward level within six months before citywide deployment.

Question 4: Is the project's purpose and basic functionality easily understood?

Answer: Yes. SmartRoad BD is a mobile and web platform that allows any citizen to photograph a damaged road and submit it with one tap. The platform's AI analyzes the photo, classifies the damage, and scores its severity automatically. Nearby citizens are invited to confirm the report. The government's authority dashboard displays all confirmed issues ranked by priority, and officials assign repair tasks to field engineers digitally. Engineers navigate to the site, perform the repair, and submit before-and-after photos for AI verification. The reporting citizen then receives a notification that their issue has been resolved. This end-to-end loop — from citizen report to verified repair to closed feedback — is the core purpose of the system, and it is intentionally designed to be understood in a single sentence: SmartRoad BD connects citizens who see road damage with the government teams who fix it, using AI to ensure that the right roads get repaired first and that every repair is verified before it is counted as done.